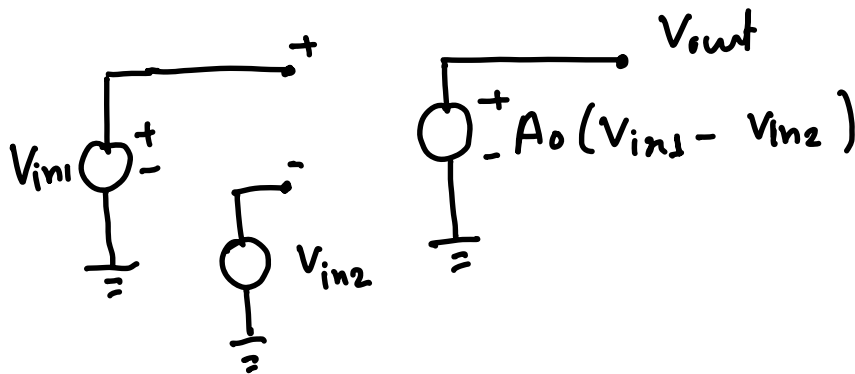
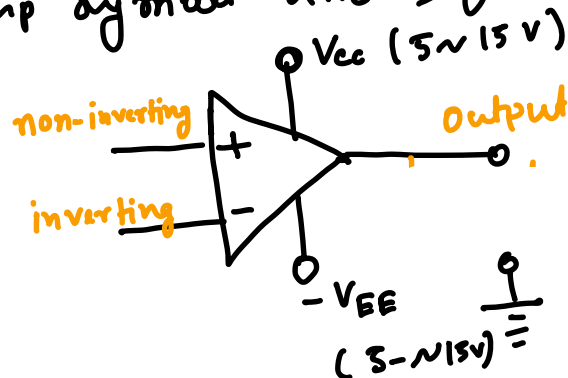


Operational Amplifier

OP-amp

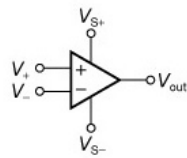
↳ each amp implemented a mathematical operation (e.g. integration), the term "operational amplifier" was born.

Op-amp symbol and equivalent circuit

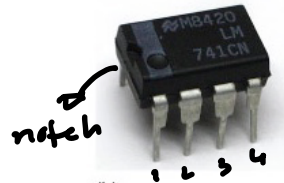


Ideal characteristics (How does the "ideal" op amp behave?)

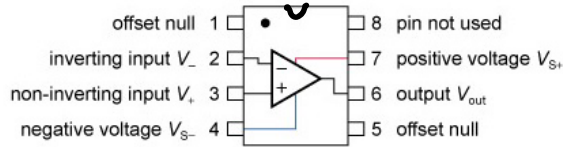
1. input resistance is very high. ($\approx \infty$)
2. Gain A_o is very high ($\approx 10^4$)
3. output resistance very small.
4. Infinite speed.



(a)



(b)



(c)

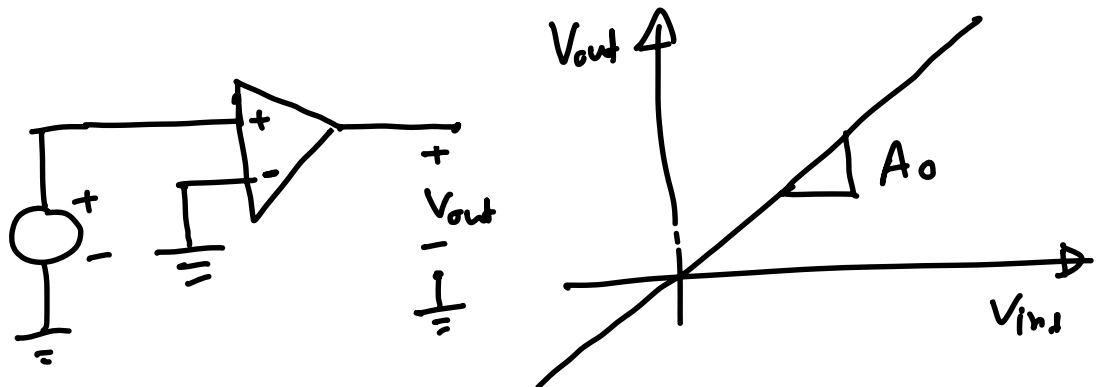
$V_{in1} \rightarrow$ non inverting input

$V_{in2} \rightarrow$ Inverting input

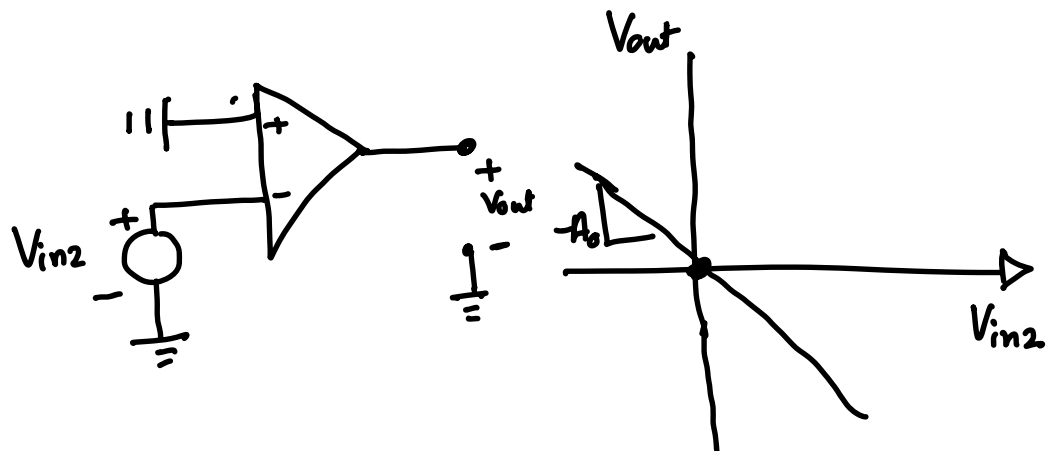
$$V_{out} = A_o (V_{in1} - V_{in2})$$

$\hookrightarrow$ open-loop gain

non-inverting characterization



Inverting characteristics



high gain ($\approx \infty$) of the op amp leads to an important observation

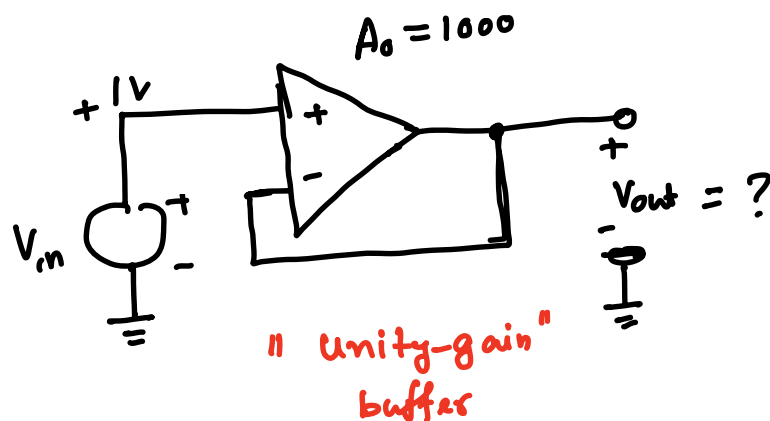
$$V_{in} - V_{in2} = \frac{V_{out}}{A_0}$$

$$\Rightarrow V_{in} - V_{in2} = \frac{V_{out}}{\infty} \rightarrow 0$$

$$\Rightarrow V_{in1} = V_{in2} \quad (A = \infty)$$

Common mode rejection

Example 8.1



$$V_{out} = A_o (v_{in1} - v_{in2})$$

$$= A_o (v_{in} - v_{out})$$

$$\Rightarrow v_{out} = A_o v_{in} - A_o v_{out}$$

$$\Rightarrow A_o v_{in} = v_{out} (1 + A_o)$$

$$\Rightarrow v_{out} = \frac{A_o}{1 + A_o} v_{in}$$

$$= \frac{1000}{1 + 1000} v_{in}$$

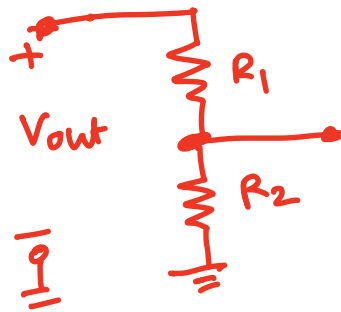
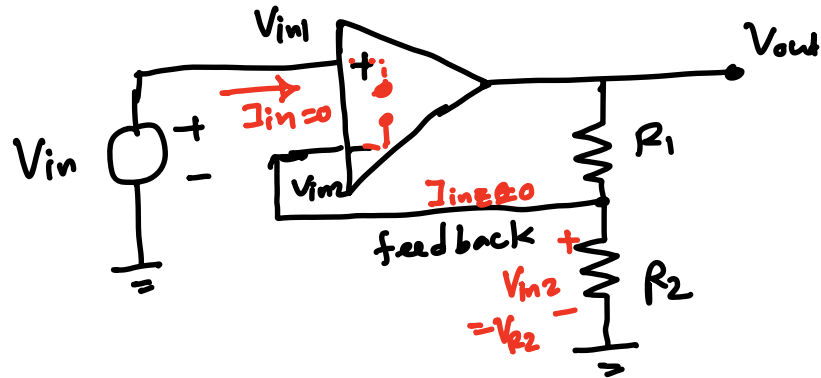
$$= \frac{1000}{\cancel{1000} \cdot 1000} v_{in}$$

$$\therefore v_{out} = v_{in}$$

$$\therefore \frac{v_{out}}{v_{in}} = A_v = 1$$

Op-amp based circuits

Non-inverting Amplifier



$$V_{R2} = \frac{V_{out} \times R_2}{R_1 + R_2}$$
$$= V_{in2}$$

A_v is very high

$$V_{in1} \approx V_{in2}$$
$$V_{in} = \frac{V_{out} \times R_2}{R_1 + R_2}$$

$$\therefore V_{in} = \frac{V_{out} \times R_2}{R_1 + R_2}$$

$$\Rightarrow V_{in} (R_1 + R_2) = V_{out} \times R_2$$

$$\Rightarrow V_{in} R_1 + V_{in} R_2 = V_{out} \times R_2$$

$$\Rightarrow V_{in} \frac{R_1}{R_2} + V_{in} = V_{out}$$

$$\Rightarrow V_{in} \left(1 + \frac{R_1}{R_2} \right) = V_{out}$$

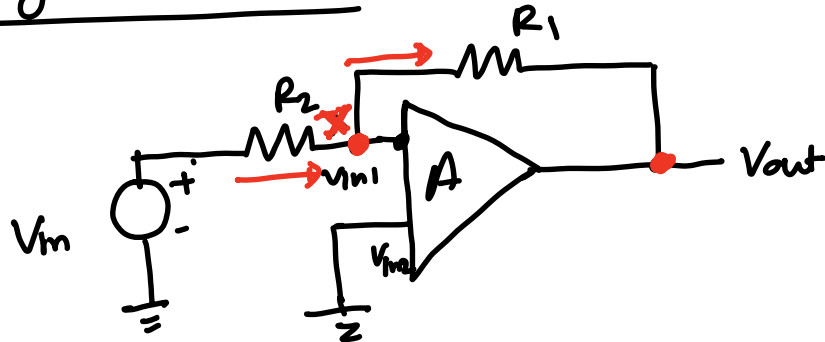
$$\Rightarrow V_{out} = V_{in} \left(1 + \frac{R_1}{R_2} \right)$$

$$\Rightarrow \frac{V_{out}}{V_{in}} = 1 + \frac{R_1}{R_2}$$

$$\therefore A_v = 1 + \frac{R_1}{R_2}$$

Input impedance, $R_{in} = \infty$

Inverting Amplifier



$$A \rightarrow \infty. \quad V_{in1} \approx V_{in2}$$

$$V_{in} \approx V_{in2} = 0$$

$$\therefore V_{in} = 0$$

X bears a zero potential even though it is not shorted to ground. For this reason, node x is called a "virtual ground."

KCL @ node X.

$$\frac{V_{in} - 0}{R_2} = \frac{0 - V_{out}}{R_1}$$

$$\therefore \frac{V_{out}}{V_{in}} = - \frac{R_1}{R_2}$$

$$R_{input} = R_2$$

Example. 8.2
